

PIYUSH SHARMA

GenAI Engineer | LLM & RAG Systems | Python

New Delhi, India · 8586998460 · piyushsh0920@gmail.com · linkedin.com/in/piyush-sharma-6591171ba · github.com/piyush0920

PROFILE

Data & GenAI practitioner with hands-on experience building retrieval-augmented generation (RAG) pipelines, LLM-based applications, and evaluation frameworks using LangChain, ChromaDB, and open-source and API-based LLMs. Background in operational data analytics (rate/rater modelling, KPI reporting, reconciliation) combined with a Data Science & AI diploma covering Python, NLP, deep learning, and cloud deployment. Comfortable working across the full GenAI stack — chunking and embeddings, vector databases, prompt engineering, and structured evaluation (RAGAS) — with a strong bias for debugging real pipeline failures end-to-end.

TECHNICAL SKILLS

- | GenAI / LLM — LangChain, RAG pipeline design, prompt engineering, LLM evaluation (RAGAS, ROUGE), Ollama, HuggingFace
- | Vector Databases — ChromaDB (persistent stores, 80K+ chunk collections), embeddings (sentence-transformers)
- | Languages & Tooling — Python (pandas, NumPy, scikit-learn), SQL, Git
- | ML / NLP — Classification & regression (SVM, Decision Tree, Ridge/Lasso), CRF-based NER, time-series forecasting (ARIMA/SARIMA)
- | Cloud & Data — AWS S3/Lambda, GCP, PySpark, Power BI (DAX, Power Query)
- | Currently deepening — FastAPI, LangGraph, LlamaIndex, Agentic AI, LoRA/QLoRA fine-tuning, AWS Bedrock, Azure OpenAI (via ongoing coursework and self-study)

GENAI & MACHINE LEARNING PROJECTS

RAG Pipeline over Legal Document Corpus ↗

- | Built a full RAG pipeline (LangChain, ChromaDB, Ollama llama3.2:3b, HuggingFace sentence-transformer embeddings) over a 4-category legal corpus with 84,844 persisted chunks
- | Designed an LCEL RAG chain with anti-hallucination prompting; ran structured evaluation via RAGAS/ROUGE (faithfulness 0.71, context precision 1.0, context recall 0.75)
- | Debugged real pipeline failures end-to-end: environment/config errors, exhausted inference credits, local-model timeouts, and evaluation-library shims

Airline Passenger Demand Forecasting ↗

- | Evaluated 13 forecasting models (Naive through SARIMA) with Box-Cox transformation and RMSE/MAPE benchmarking (statsmodels, scipy)

Cardiovascular Risk Prediction ↗

- | Developed and tuned SVM and Decision Tree classifiers on 68,775 patient records with GridSearchCV, class imbalance handling, and sensitivity-specificity cutoff selection

Namma Yatri Ride-Hailing Power BI Dashboard ↗

- | Built a 5-table star schema with DAX and Power Query; delivered full EDA across demand, revenue, and funnel metrics

WORK EXPERIENCE

Senior Executive, P&C Insurance Operations (Management Liability & Professional Lines)

Xceedance · New Delhi, India · 2+ years (promoted from Executive within 2 years)

- | Own Excel-based rater tools and financial models for policy pricing; built and maintained KPI dashboards and reconciliation reporting across large operational datasets
- | Authored SOPs and trained new team members, translating complex data workflows into clear, repeatable processes for non-technical stakeholders

EDUCATION

Executive PG Diploma in Data Science & AI — IIIT Bangalore (delivered via upGrad)

Graduating 2026 · GPA: 3.82/4.00 · Coursework: Python, SQL, Statistics, ML, Deep Learning/NLP (PyTorch, TensorFlow), GenAI/LLM specialization (LangChain, prompt engineering, vector databases), Cloud (AWS/GCP), PySpark

B.A. in Mathematics and Office Management (OMSP) — University of Delhi

2020–2023 · Coursework in Advanced Mathematics, Operational Research, R Programming